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Prelim Exam

Create an ansible file:
yml:
First is installing python3 and pip3 on Manage node:

```yaml

- name: Install Python3 and pip3
  apt:
    name:
      - python3
      - python3-pip
    state: latest

```

apt for installing our packages:
python3 and python3-pip

```yaml
- name: Install Python MySQL connector
  apt:
    name: python3-pymysql
    state: present
```

Needed for Mariadb to work properly

```yaml
- name: Set python3 as default python
  alternatives:
    name: python
    path: /usr/bin/python3
```

On this part, check where does the python3 package installed, then put it on ansible playbook to use it to setup the python we installed as default especially if there's another python installed.

```yaml
- name: Install OpenJDK
  package:
    name: openjdk-21-jdk
    state: present
```

This one installing OpenJDK, it's a java packages

```yaml
- name: Install MariaDB
  package:
    name:
      - mariadb-server
      - mariadb-client
    state: present

- name: Start MariaDB Service
  service:
    name: mariadb
    state: started
    enabled: yes
```

Installing mariadb and also starting it. When we installing something on our manage node, we often use apt, yum or what. But since we're going to start the MariaDB service, this time we will use 'service:' then enabled it.

```yaml
- name: Create database
  mysql_db:
    name: prelim
    state: present
    login_user: root
    login_unix_socket: /var/run/mysqld/mysqld.sock
```

```yaml
- name: Create table and insert record
  mysql_query:
    login_db: root
    query:
      CREATE TABLE IF NOT EXISTS exam_table
      (id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(50));
      INSERT INTO exam_table (name) VALUES ('Ansible Exam');
```

This code creates database and also create table, I also add IF NOT EXISTS since i'm trying this over and over until the codes correct.

```yaml
- name: Set MOTD message
  copy:
    content: "{{ motd_message | default('Ansible Managed node by Isaac') }}"
    dest: /etc/motd
```

```yaml
- name: Create user
  user:
    name: Isaac
    state: present
```

The last part is creating message of the day or MOTD, and then create user.

ini:
This is my ip address on manage node

```
ansible.cfg   config.yml   inventory.ini
Alegre_PrelimExam > inventory.ini
[Isaac]
192.168.56.114

[Jhon]
192.168.56.115
```

cfg:
This is the ansible config of my project, where we can put our private key location and also our inventory

```
legre_PrelimExam > ansible.cfg
[defaults]
inventory = inventory.ini
private_key_file = /home/isaac/.ssh/id_rsa
```

Results:

```
ansible.cfg    ! config.yml    inventory.ini
Alegre_PrelimExam > ! config.yml
  2   - name: Prelim Playbook
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS          bash - Alegre_PrelimExam

isaac@Workstation:~/ansible/Alegre_PrelimExam$ ansible-playbook config.yml -K

TASK [Install Python3 and pip3] ***********************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Install Python MySQL connector] *****************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Install OpenJDK] ********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Install MariaDB] ********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Start MariaDB Service] **************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Create database] ********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Set MOTD message] *******************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [Create user] ************************************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.114]
```

```
TASK [Set python3 as default python] ******************************************************************************
fatal: [192.168.56.114]: FAILED! => {"changed": false, "msg": "Needed to install the alternative, but unable to do so as we are missing the
link"}
fatal: [192.168.56.115]: FAILED! => {"changed": false, "msg": "Needed to install the alternative, but unable to do so as we are missing the
link"}

PLAY RECAP ********************************************************************************************************
192.168.56.114             : ok=9    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
192.168.56.115             : ok=9    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

isaac@Workstation:~/ansible/Alegre_PrelimExam$
0 0                                                   Ln 61, Col 7   Spaces: 2   UTF-8   LF   { } YAML
```

Alternative is missing, idk why

Verify:
Python3 and pip3

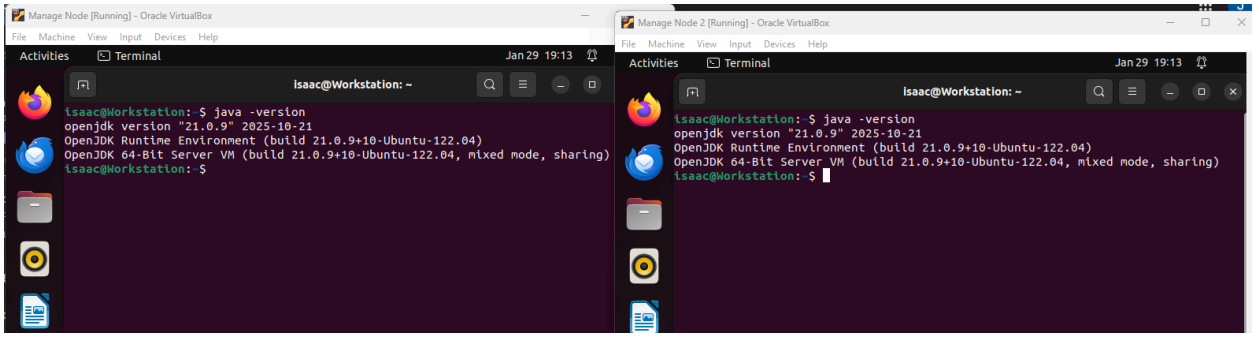
```
Activities   Terminal                                   Jan 29 19:12
                                   isaac@Workstation: ~
isaac@Workstation:~$ python3 & pip3 --version
[1] 16416
Python 3.10.12 (main, Jan  8 2026, 06:52:19) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)

[1]+  Stopped                 python3
isaac@Workstation:~$
```

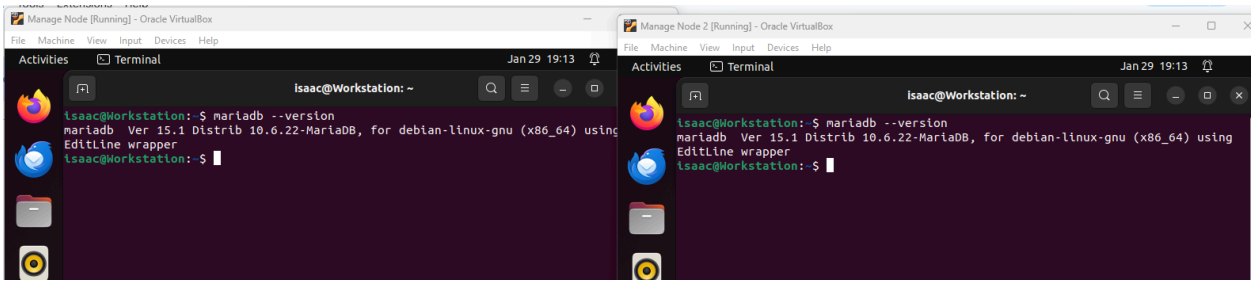
```
Activities   Terminal                                   Jan 29 19:12
                                   isaac@Workstation: ~
isaac@Workstation:~$ python3 & pip3 --version
[1] 16960
Python 3.10.12 (main, Jan  8 2026, 06:52:19) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)

[1]+  Stopped                 python3
isaac@Workstation:~$
```

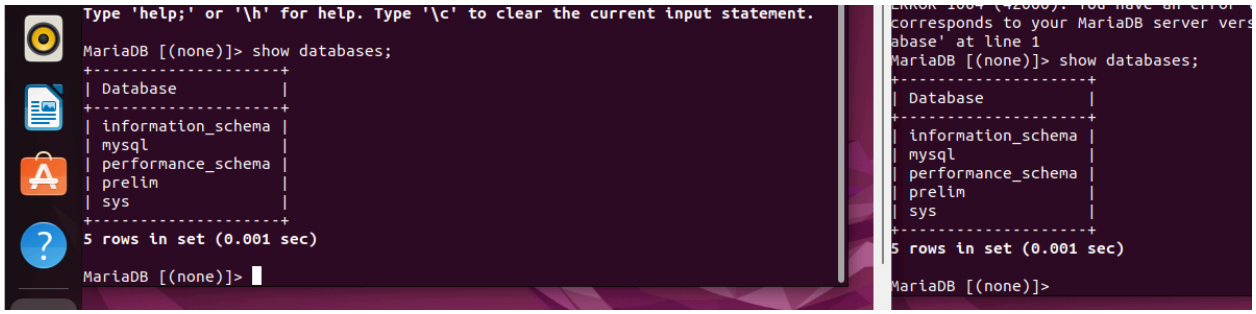
Openjdk



Mariadb



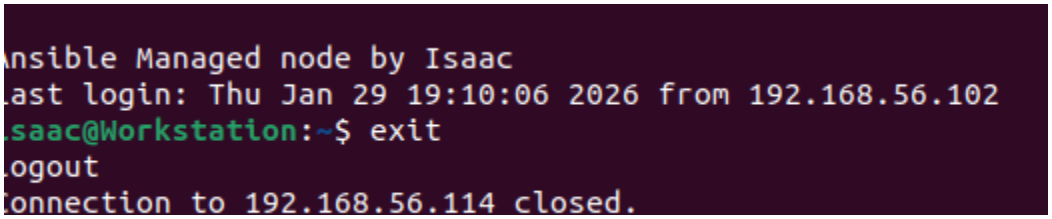
Database



Table

Wrong syntax

User for .114



User for .115

```
Ansible Managed node by Isaac
Last login: Thu Jan 29 19:10:06 2026 from 192.168.56.102
isaac@Workstation:~$ exit
logout
Connection to 192.168.56.115 closed.
isaac@Workstation:~$
```